

Assignment-3

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Q. To understand DevOps: Principles, Practices and DevOps Engineer Role and Responsibilities.

DevOps is a combination of Development (Dev) and Operations (Ops). It is a culture, set of practices, and tools that improves collaboration between software developers and IT operations team. The main goal of DevOps is to deliver software faster, with better quality, and with continuous improvement.

* Principles of DevOps:

1. Collaboration and Communication:

Development, testing, and operations teams work together throughout the software lifecycle.

2. Automation:

Repetitive tasks like testing, building, deployment and monitoring are automated to save time and reduce errors.

3. Continuous Integration (CI):

Developers frequently merge code into a shared repository where builds and tests run automatically.

4. Continuous Delivery / Continuous Deployment (CD):

Software updates are released quickly and reliably through automated pipelines.

5. Monitoring and Feedback:

Applications and infrastructure are continuously monitored to detect issues and improve performance.

6. Customer centric approach:

Focus is on delivering value to users faster with frequent updates and improvement.

7. Infrastructure as code (IaC):

Servers, networks and environments are managed using code instead of manual configuration.

* DevOps Practices:

- Continuous Integration (CI)
- Continuous Delivery (CD)
- Continuous Deployment
- Version Control
- Automated Testing
- Configuration management
- Containerization
- orchestration
- monitoring and logging.

* Role of a DevOps Engineers

A DevOps Engineer is responsible for bridging the gap between development and operations teams. They automate workflows, manage infrastructure and

ensure reliable software delivery.

They work with developers, testers, security teams and cloud teams to maintain smooth release cycles.

* Responsibilities of a DevOps Engineer:

- Build and Release management: create automated build and deployment pipelines.
- CI/CD pipeline setup: use tools like Jenkins, Github Actions or Github CI/CD.
- Infrastructure management: provision and maintain servers, cloud resources and environments.
- Automation: automate testing, deployments, backups and repetitive tasks.
- monitoring and Incident response:
- Security Integration (DevSecOps): Include security checks in pipelines and infrastructure.
- Collaboration: coordinate between developers, QA, and operations teams.
- Performance optimization: improve scalability, reliability, etc.
- Documentation: maintain runbooks, deployment guides, and architecture documents.